Optimization design of nuclear power plant monitoring display interface based on SEEV model

Rui Xing
School of Mechanical Engineering
Southeast University
Nanjing, Jiangsu, China
220224491@seu.edu.cn

Jinzhen Dou
School of Mechanical Engineering
Southeast University
Nanjing, Jiangsu, China
230208500@seu.edu.cn

Chengqi Xue*
School of Mechanical Engineering
Southeast University
Nanjing, Jiangsu, China
ipd_xcq@seu.edu.cn

Abstract

Based on the SEEV model, the nuclear power plant monitoring and display interface is optimized and designed. First, we summarize the key contents and existing problems of the interface, then introduce the salience and effort factors in the SEEV model to optimize the design, and finally use eye tracking methods to study the old and new interfaces. Combined with the analysis of eye movement experimental data, the visual cognitive characteristics of subjects performing visual search tasks in different interfaces were obtained. Experimental results show that combining the SEEV model can better guide subjects' attention allocation strategies, thereby improving work efficiency.

CCS Concepts

• **Human-centered computing** → Interaction design; Interaction design process and methods.

Keywords

HMI, SEEV model, Nuclear power plant monitoring and display interface, Eye movement experiment

ACM Reference Format:

Rui Xing, Jinzhen Dou, and Chengqi Xue. 2024. Optimization design of nuclear power plant monitoring display interface based on SEEV model. In *2024 5th International Conference on Computer Science and Management Technology (ICCSMT 2024), October 18–20, 2024, Xiamen, China.* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3708036.3708038

1 Introduction

In recent years, with the continuous progress of science and technology and the enhancement of information level, the nuclear energy industry has also made remarkable progress in the design of monitoring display interface. As a core tool for operation and management, the interface design of nuclear power plant monitoring system is not only related to the efficiency of operation personnel and the safety of the plant, but also directly affects the production and operation of the plant, so it should comprehensively consider the reliability, convenience, comfort and so on[1].

The current nuclear power plant monitoring and display interface is dominated by the digital human-machine interface, which, while realizing the centralization, automation and precision of system management and control, also brings the problem of limited display of a huge amount of information, and increases the source of new human error[2]. Excessive information presentation may cause the manipulator to enter a complex cognitive phase, resulting in cognitive difficulties such as operational errors, misinterpretation and misjudgment, untimely feedback, etc. thus leading to mission failure, and in serious cases, even causing major accidents[3,4]. Attentional resources are valuable resources in the human cognitive system, directly affecting the ability of operators to process information, make decisions and respond to unexpected situations, such as SEEV Model[5]. HORREY et al[6] applied the SEEV model to the driving field to study the effects of task-related information bandwidth, task priority, and traffic hazards on driver scanning behavior, and the accuracy of the SEEV model in predicting scanning behavior is demonstrated. Wortelen et al[7] extended SEEV model to the Adaptive Information Expectancy (AIE) model, which is used to create the driver's dynamic cognition. Xu Juanfang[8] took the factors of the SEEV model as a starting point, translated it into a design strategy for automotive instrument panels by combining the laws of eye gaze.

According to the aforementioned literature, the academic and scientific validity of the SEEV model has been widely verified in the field of attention allocation prediction research. However, its current applications are mainly focused on aerospace and driving domains, and it is rarely applied to nuclear power plant monitoring scenarios. Therefore, this study aims to explore the optimization design of nuclear power plant monitoring display interface based on the SEEV model, so as to provide more intuitive and efficient monitoring display interface for nuclear power plants, and to improve the efficiency of the operating personnel and the operation and management level of the nuclear power plants.

2 Related theory

2.1 Monitor Display Interface Overview

2.1.1 Screen content. The following control tasks are typical in nuclear power plants: switching; dynamic control; inspection and testing; supervisory control[9]. In the monitoring display interface, the process screen is an important aspect of the human-machine interface design, and is the main medium of information exchange between the operator and the instrument and control system. Taking the Hualong One chemical and volumetric control system (RCV)

*Corresponding author.

ICCSMT 2024, October 18–20, 2024, Xiamen, China

ACM ISBN 979-8-4007-0999-9/24/10
https://doi.org/10.1145/3708036.3708038

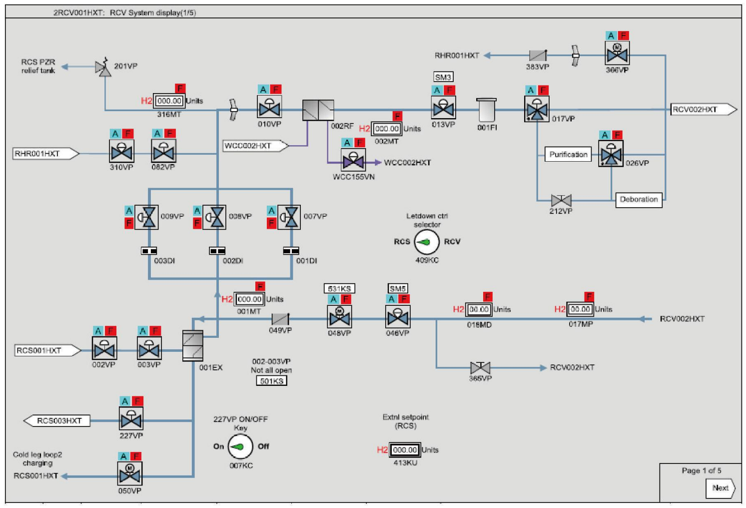

Figure 1: Hualong One chemical and volumetric control system (RCV) interface.

interface[10] as an example (Figure 1), the process interface mainly contains operational information (e.g., dynamically linked labels and valves) and non-operational information (e.g., equipment and piping), including information on piping, auxiliary sprinklers, and the level of the capacitance control tanks and voltage stabilizers, etc.

2.1.2 Existing issues. The process flow chart screen is based on the nuclear power system flow chart to display the simplified design of the process flow, can send commands to the process and display operational feedback information, mainly used to perform the operational tasks in the protocol. There are three main problems in the current process flow screen:

(1) The results of the WCAG (Web Content Accessibility Guidelines) contrast test show that several color combinations for the current interface do not meet the standard, for example, the contrast between the background color of the screen and the color of some interface elements in the current interface is not significant. is not a significant contrast effect.

(2) In terms of layout, due to the large amount of information and complex structure of the monitoring and display system of a nuclear power plant, the interface layout is too compact, resulting in unclear information structure, information repetition, and information redundancy.

(3) In the current interface, the layout of text and other elements is too compact, and the choice of font size does not reflect the hierarchy of information. At the same time, the differences between different graphical symbols are small, and the design of the existing symbols also has some room for improvement in terms of aesthetics. In addition, most of the process lines in the process flow diagram are thin and not obvious enough to reflect the hierarchical relationship of the interface elements.

2.2 SEEV model

Factors affecting attention are multifaceted, and there are two main factors: (1) an individual's motivation or needs, in the presence of motivation and needs, special attention will be paid to factors that fulfill the needs; (2) from the stimulus itself.

The SEEV model is influenced by four factors, namely, information salience, effort, expectancy, and value, and establishes an attention allocation relationship by linearly weighting the four factors. Salience is a bottom-up factor that highlights features within the area of interest (AOI) that naturally attract attention. Effort is another bottom-up factor that involves the physical movements required to gather information. Expectation combines the frequency of information updates within the AOI with the user's expectation of these changes. Value relates to the importance of the AOI relative to the main goal of the task.

3 Optimized design of monitoring display interface

3.1 Design based on salience

3.1.1 Color and contrast. The use of color and contrast is key in the design of nuclear power plant monitoring display interfaces. Studies have shown that low saturation/lightness coding is appropriate for backgrounds or large color blocks; high saturation/lightness coding is appropriate for interface elements or small color blocks, and color saturation should not be too high for persistent or non-dynamic secondary information[11,12,13]. Therefore, the background color is set to F2F2F2, and the text color is set to 000000. the selection of functional colors should be in line with the relevant nuclear power interface design principles and meet the design conventions of the nuclear power industry.

3.1.2 Layout and space allocation. Layout design principle refers to the fact that the human-machine interface of nuclear power main control room should be based on tasks or modules, and the interface display and control objects should be categorized and arranged[14], so that the manipulator can quickly access interface information, efficiently conduct visual search and smoothly complete the relevant operations.

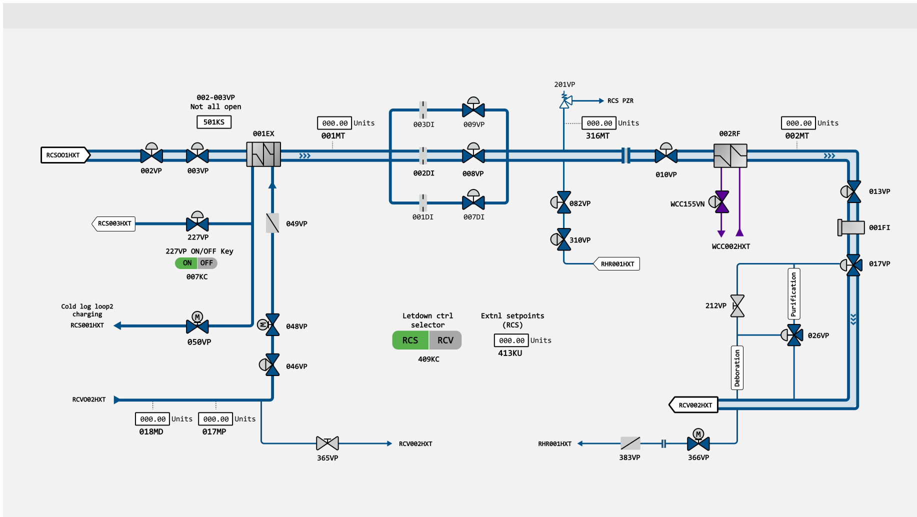

Figure 2: Optimized interface based on salience.

An effective layout should take into account the visual habits of the operator and place the most important information where it is most easily noticed. According to a Nielsen survey, people's visual trajectory when reading follows an "F" shape. Therefore, the most important information should be placed at the top and at the beginning of the line to attract the operator's attention[15]. In addition, while ensuring the readability of the process flow diagram, the bending and crossing of pipelines are simplified as much as possible to reduce the complexity of the interface; the labeling of the control and status information of some equipments should be clear, and follow the principle of proximity of the format tower to enhance the readability of the interface[16]; meanwhile, the pointing of the dynamic link labels is consistent with the semantics of the flow direction of the pipelines.

3.1.3 Text, symbols and line design. Font selection should consider clarity and legibility. Sans-serif fonts are usually more suitable for digital displays, as their clean lines are helpful for quick reading. Therefore, Consolas and Bold, which are common standard fonts, are chosen, and the font size is set to 20 for the first level and 16 for the second level. in terms of color selection, after the WCAG color contrast test, the contrast between the text color and the background color is 8.09, which is in line with the standard.

The overall style of the symbol design is unified as a minimalist style. The shapes are basically retained and inherited from the original, while the rounded corners are larger compared to the original symbols, giving a softer feeling and reducing the sense of visual oppression. In addition, some symbols, such as valves and gates in the design of the pipeline into the pipeline, can visually represent the pipeline medium flow and disconnection.

The arrows in the process line are designed in accordance with the equipment pipeline for pointing to the direction of media transmission, and the speed of the arrow kinetic effect shows the media flow rate. According to the level of functional hierarchy, the main line is set as eye-catching blue double line thick line (6pt) plus dynamic arrows, the main line is set as single line thick line, and the branch line is set as single line thin line (3pt).

The interface display is shown in Figure 2.

3.2 Design based on effort

3.2.1 Navigation bar. In order to reduce the operator's difficulty in searching the interface and to minimize the search time, a navigation bar is set up on the left side of the interface. The navigation bar integrates important symbols and information such as equipment, valves, data, etc., and indicates the existence of problems and normal status through red and green dots respectively.

3.2.2 Related highlighting tips. In the interface, when the operator hovers the mouse over an element, other elements associated with that element will be presented in the interface in the form of lighting up. In the navigation bar, when the mouse hovers over an icon, the same icon in the right interface will also light up to assist novice operators to quickly find the corresponding equipment and data in the flowchart.

The interfaces are shown in Figure 2 and Figure 3.

4 Experiment

This study investigated the effects of an attention allocation strategy based on saliency and effort on visual search performance using an interface form of a nuclear power process screen. The experimental hypothesis:Subjects will have better search performance for nuclear power interfaces that combine saliency and effort when compared to the original picture.

4.1 Participants

The experiment involved 22 subjects, including 12 males and 10 females, aged between 21 and 28 years. All participants were graduate students from Southeast University. To ensure the accuracy of eye movement data collection, participants had normal color vision,

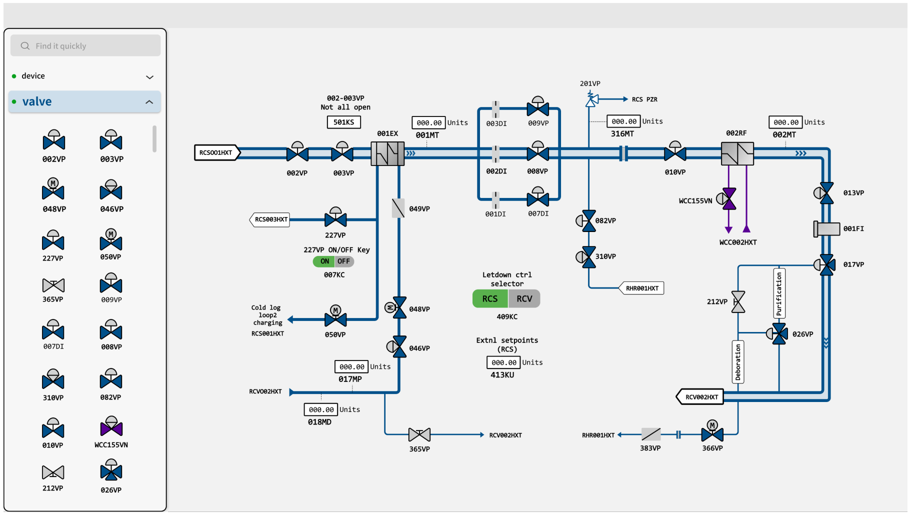

Figure 3: Optimized interface based on effort.

no astigmatism, and binocular vision of at least 4.2. Their uncorrected or corrected vision was normal. A total of 22 experiments were conducted, and 2 sets of invalid data were removed, resulting in 20 valid experiments.

4.2 Equipment and Environment

This study used Tobii Glasses 3 eye-tracking glasses, and the experiment was conducted in an environment free from external interference. The experimental setup included a 15.6-inch screen with a resolution of 1920×1080. Participants sat in an adjustable chair facing the screen, and the eye tracker sampled eye movement information at a frequency of 120Hz. The experiment was programmed using Tobii Pro Lab, and the stimuli were displayed on the computer screen.

4.3 Experimental Design

4.3.1 Experimental variables and materials. The independent variable was the design of the nuclear power process interface. The control group used the original nuclear power process interface (Figure 1), the experimental group 1 used a salience-dominant interface (Figure 2), and the experimental group 2 used an interface integrating salience and effort (Figure 3).

4.3.2 Experimental procedure. Participants filled out personal information questionnaires and informed consent forms, were briefed on the experimental steps, and practiced to familiarize themselves with the procedure. Calibration was performed before the experiment. Participants were instructed to maintain eye height at the center of the screen, adjust the screen distance to 40-60 cm, and keep their body and head still during the experiment. After calibration, the calibration effect was tested. Participants read the instructions, pressed any key to start practice trials, and then began the formal experiment.

At the beginning of the experiment, participants would first see the gaze point "+" in the center of the screen with a duration of 1000 ms, and then enter the task interface after 1000 ms to find the target icon (divided into valves and data frames, which appeared randomly) and respond by pressing the numeric keys. After the search was completed, the next round of search trials was conducted, and the trials had a total of 18 (3×6) search tasks. During the process of searching for the target shape, subjects were required to find the first target and respond to it before proceeding to the next trial, as shown in Figure 4.

Table 1: ANOVA results.

Figure	Index	μ	N	sd	se
1	Total fixation time	58.57	20	19.54	4.37
	Fixation count	107.60	20	47.27	10.57
2	Total fixation time	47.77	20	11.76	2.63
	Fixation count	92.50	20	39.95	8.93
3	Total fixation time	44.63	20	8.97	2.01
	Fixation count	79.95	20	33.48	7.49

4.4 Experimental Results

4.4.1 Total fixation time. Total fixation time indicates the duration of gaze within the screen, reflecting the cognitive load. Data showed significant differences in total fixation time among different groups, as is shown in Table 1 and Figure 5. The total fixation time for Figure 1 and Figure 2 was 58.57 seconds and 47.77 seconds, respectively, while for Figure 3, it was 44.63 seconds. Analysis of variance results indicated significant differences in total fixation time ($F=5.343$, $p=0.00747$, $p< 0.01$). This suggests that the optimized interface design reduced visual search reaction time, validating the advantages of salience and effort-based design in reducing cognitive load and improving operational efficiency.

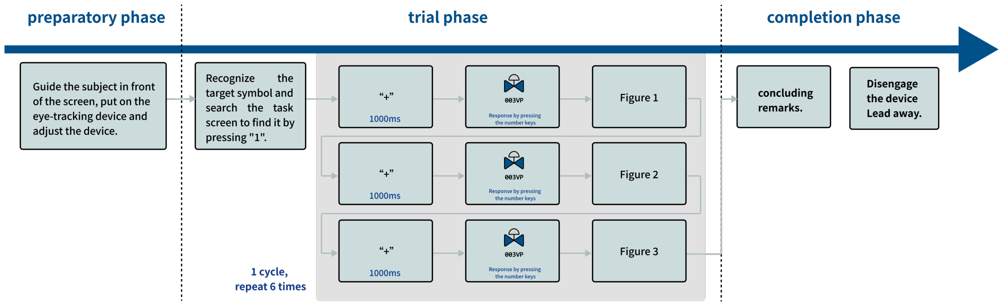

Figure 4: Experimental procedure.

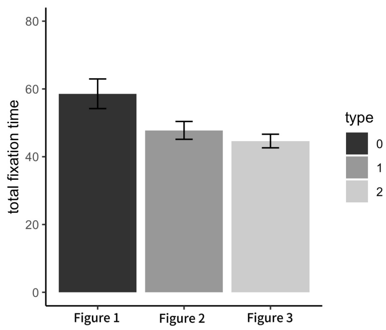

Figure 5: Average total fixation time.

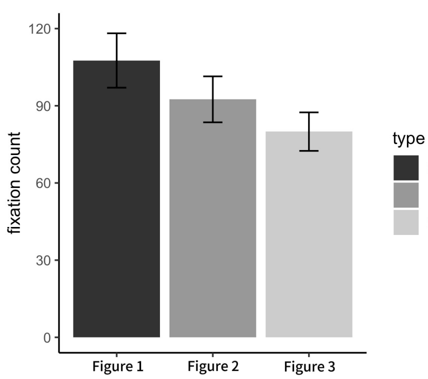

Figure 6: Average fixation count.

4.4.2 Fixation count. Fixation count reflects the frequency of information search and confirmation. Data in Table 1 and Figure 6 showed that the average fixation count for Figure 3 was significantly lower than for other images (Figure 1 107.60 times, Figure 2 92.50 times, Figure 3 79.95 times). This result suggests that optimized interface design led to fewer fixations and reduced information search burden.

4.4.3 Heat map. Eye-tracking heat maps show user browsing and fixation patterns. Red areas indicate concentrated gaze regions, while yellow and green areas indicate less attention.

The heat maps of the three pictures were generated using the software for further analysis, and the specific results are shown in Figure 7 and Figure 8 below. As can be seen from the heat maps, in both Figure 7 and Figure 8, the subjects showed a relatively even interest in the pictograms in the interface, with a more dispersed gaze point and a relatively large green range in the eye-movement hotspot. The distribution of subjects' attention to the interface was more dispersed, and no significant attention was allocated to the target icon part of the visual search. Therefore, the interface has the problem that the non-visual attention focus area of the subjects is large, which not only underutilizes the information of the interface, but also is not conducive to the subjects' recognition of important information.

Figure 9 shows the eye-movement heat map of Figure 3, reflecting the user's browsing and gaze on the interface. As can be seen from the figure, in the whole interface, the subjects showed higher interest in the information in the navigation bar area, with denser attention points and more red eye-movement hotspots; whereas the right flow interface module was basically unattended by the subjects, with hotspots focusing mainly on the target pictograms. This suggests that in this interface, the visual attention points and visual flow of the manipulator are simpler, focusing only on specific areas in the process flow diagram section, and locating the corresponding equipment and making operations in a shorter time.

5 Conclusion

Based on the SEEV model, this study explores the optimal design of the monitoring display interface in nuclear power plants. By analyzing the effects of different interface designs on the operator's eye movement behavior, three conclusions are drawn: (1) Optimizing interface color, layout and visual elements can improve the saliency of the interface, and the total gaze time and the number of gaze times of highly salient interfaces are significantly lower than those of other interfaces, which can effectively reduce the cognitive load of the operator and improve the operation efficiency. (2) The design of navigation bar and associated highlighting hints can reduce the cost of operators' efforts to obtain information, reduce the total gaze time and the number of gaze times, and enable operators to quickly locate and obtain information. (3) The analysis of the heat map shows that the operator's expectation and assessment of the importance of the information will affect the allocation of attention and interface interaction behavior, so the interface design should be based on the operator's subjective expectations, optimize the

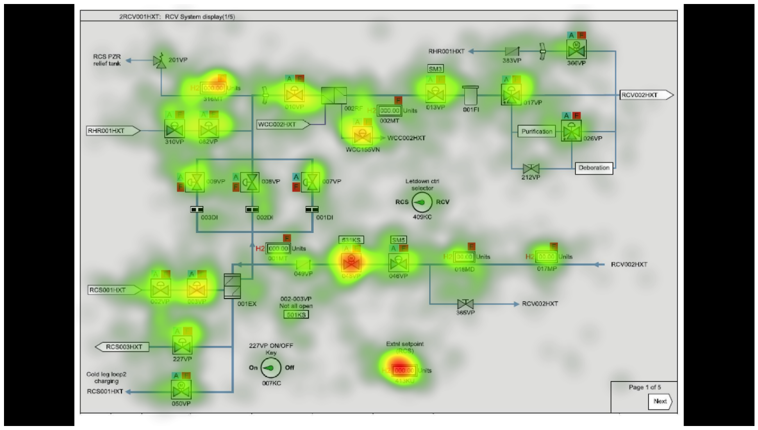

Figure 7: Original interface heat map.

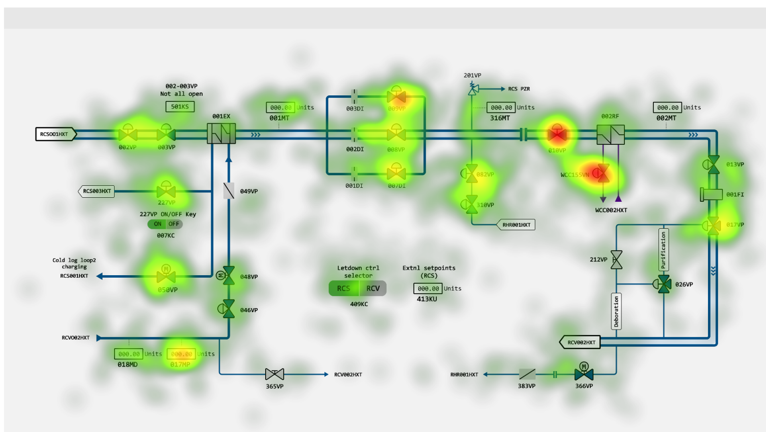

Figure 8: Optimized interface based on salience heat map.

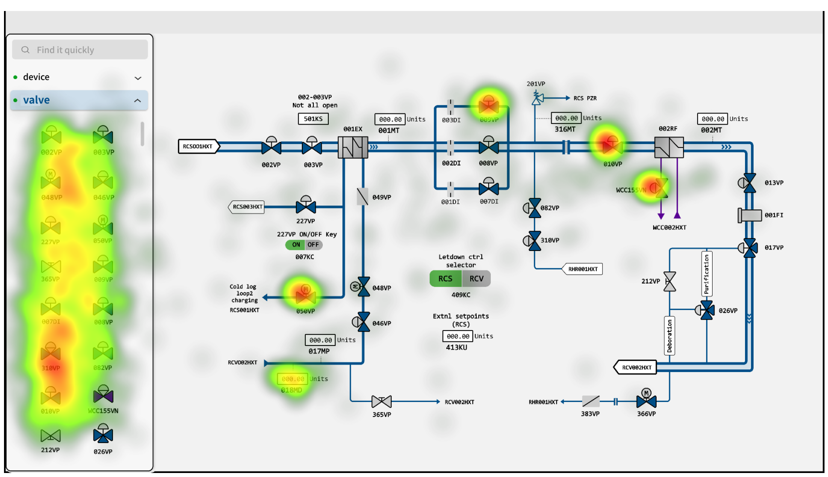

Figure 9: Optimized interface based on effort heat map.

presentation and layout of information, and ensure that the prominence and accessibility of high-value information are more in line with the operator's operating habits and experience.

However, this study still has some limitations. The singularity of the experimental task and the limited sample size may lead to insufficient generalization ability of the results. In addition, the complexity and individual operator differences in the actual operating environment need to be further considered in future studies. Future studies should expand the sample size, design more diverse experimental tasks, conduct long-term effects studies, and collect data in real or more nearly real environments. Considering physiological, psychological, and behavioral data in an integrated manner and conducting a multi-perspective analysis will help to gain a comprehensive understanding of the operator's needs and behavioral characteristics, which will further optimize the interface design and ensure the safe and efficient operation of the nuclear power plant.

References

[1] Lagari, P. L., Nasiakou, A., Fainti, R., Mao, K., Tsoukalas, L. H., Bean, R., & Alamaniotis, M. 2016, July. Evaluation of human machine interface (HMI) in nuclear power plants with fuzzy logic method. In 2016 7th International Conference on Information, Intelligence, Systems & Applications (IISA) (pp. 1-6). IEEE.

[2] Zhang, L. 2010. The effect of information display on human reliability in a digital control room. Zhongguo Anquan Kexue Xuebao, 20(9), 81-85.

[3] Niu Yafeng, Wu Wenyu, Zhou Lei, Zhou Xiaozhou, Xue Chengqi, & Wang Haiyan. 2018. Prospects of interactive electronic technology manuals integrating virtual reality technology in mechanical drawing teaching. Mechanical Design, (S2), 51-53.

[4] Li Jing, Xue Chengqi, Wang Haiyan, Zhou Lei, & Niu Yafeng. 2013. Human-computer interface information encoding with balanced time pressure. Journal of Computer-Aided Design and Graphics, 25(7), 1022-1028.

[5] Wickens, C. D., Helleberg, J., Goh, J., Xu, X., & Horrey, W. J. 2001. Pilot task management: Testing an attentional expected value model of visual scanning. Savoy, IL, UIUC Institute of Aviation Technical Report.

[6] Horrey W J, Wickens C D, Consalus K P. Modeling drivers' visual attention allocation while interacting with in-vehicle technologies[J]. Journal of Experimental Psychology: Applied, 2006, 12(2): 67.

[7] Wortelen, B., Baumann, M., & Lüdtke, A. 2013. Dynamic simulation and prediction of drivers' attention distribution. Transportation research part F: traffic psychology and behaviour, 21, 278-294.

[8] Xu Juanfang. 2016. Automobile instrument interface design based on SEEV attention model. Mechanical Design, 33(12), 115-118.

[9] Anokhin, A., & Chernyaev, A. 2018, August. Adaptive Human–Machine Interface Supporting Operator's Cognitive Activity in Process Control Systems. In Congress of the International Ergonomics Association (pp. 733-742). Cham: Springer International Publishing.

[10] Chu Xueqin, Cui Zepeng, & Li Yong. 2021. Design and implementation of screen elements for nuclear power plant digital control system. Nuclear Electronics and Detection Technology, 41(3), 484-490.

[11] Hou Yanhong, Zhang Lin, & Miao Danmin. 2008. Study on the physiological and performance effects of color background on visual cognitive tasks. Chinese Journal of Clinical Psychology, 16(5), 506-508.

[12] Zhang Lei, Zhuang Damin, & Wanyan Xiaoru. 2009. Information color coding based on different mental loads and task types. Journal of Ordnance Industry, (11), 1522-1526.

[13] Kahneman, D. 1973. Attention and effort (Vol. 1063, pp. 218-226). Englewood Cliffs, NJ: Prentice-Hall.

[14] Li Leshan. 2004. Human-computer interface design. Science Press.

[15] Nielsen, J. 2006. The 90-9-1 Rule for Participation Inequality in Social Media and Online Communities https://www. nngroup. com/articles/participation-inequality. Accessed Oct, 9, 2017.

[16] Zeng, L. 2009. Designing the User Interface: Strategies for Effective Human-Computer Interaction by B. Shneiderman and C. Plaisant: Pearson Addison-Wesley, 2009. xviii+ 606 pages. ISBN: 978-0-321-53735-5.